

Prueba parcial U2

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Link de repositorio en Github:

https://github.com/DianyGuerra/PruebaParcialMovilesU2_GuerraDiana.git

API:

<https://thecatapi.com/>

Con la versión gratuita de la API solo se pueden obtener hasta 10 gatitos

Endpoint para obtener gatitos

<https://api.thecatapi.com/v1/images/search?limit=10>

Consumo de la API

```

lib > data > datasource > cat_api_datasource.dart > ...
1  import 'package:http/http.dart' as http;
2  import 'dart:convert';
3  import '../models/cat_model.dart';
4
5  class CatApiDataSource {
6    final String baseUrl = "https://api.thecatapi.com/v1/images/search";
7
8    Future<List<CatModel>> fetchCats(int limit) async {
9      final url = Uri.parse("$baseUrl?limit=$limit");
10
11      final res = await http.get(url);
12
13      if (res.statusCode != 200) {
14        throw Exception("Error al cargar los gatos");
15      }
16
17      final data = json.decode(res.body);
18      final List results = data;
19      return results.map((e) => CatModel.fromJson(e)).toList();
20    }
21  }
22

```

Cat_repository

```

lib > data > repositories > cat_repository_implementation.dart > ...
1  import '../domain/entities/cat.dart';
2  import '../datasource/cat_api_datasource.dart';
3
4  class CatRepositoryImplementation {
5    final CatApiDataSource datasource;
6
7    CatRepositoryImplementation(this.datasource);
8
9    Future<List<Cat>> getCats({int limit = 10}) async {
10      return datasource.fetchCats(limit);
11    }
12  }
13

```

Modelo de gatitos

lib > domain > entities > cat.dart > Cat

```
1 class Cat {
2   final String id;
3   final String imageUrl;
4   final int width;
5   final int height;
6
7   Cat({
8     required this.id,
9     required this.imageUrl,
10    required this.width,
11    required this.height,
12  });
13 }
14
```

Conversión de JSON a objeto Dart

lib > data > models > cat_model.dart > ...

```
1 import '../domain/entities/cat.dart';
2
3 class CatModel extends Cat {
4   CatModel({
5     required super.id,
6     required super.imageUrl,
7     required super.width,
8     required super.height,
9   });
10
11   // Convertir de JSON a objeto
12   factory CatModel.fromJson(Map<String, dynamic> json) {
13     return CatModel(
14       id: json["id"],
15       imageUrl: json["url"],
16       width: json["width"],
17       height: json["height"],
18     );
19   }
20 }
21
```

ViewModel de gatitos para gestionar el estado

```
lib > presentation > viewmodels > cat_viewmodel.dart > ...
1  import 'package:flutter/material.dart';
2  import '../domain/entities/cat.dart';
3  import '../domain/usecases/get_cats_usecase.dart';
4
5  class CatViewModel extends ChangeNotifier {
6    final GetCatsUseCase getCatsUseCase;
7
8    CatViewModel(this.getCatsUseCase);
9
10   // Manejo del estado
11   List<Cat> cats = [];
12   bool loading = false;
13   String? errorMessage;
14
15   Future<void> loadCats() async {
16     loading = true;
17     notifyListeners();
18     try {
19       cats = await getCatsUseCase();
20     } catch (e) {
21       errorMessage = "Error al cargar los datos";
22     }
23     loading = false;
24     notifyListeners();
25   }
26 }
27
```

HomePage

lib > presentation > views > home_page.dart > HomePageState

```
12  class HomePageState extends State<HomePage> {
24    Widget build(BuildContext context) {
30
37      return Scaffold(
38        appBar: AppBar(title: Text('Gatitos')),
39        body: GridView.builder(
40          padding: EdgeInsets.all(8.0),
41          gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
42            crossAxisCount: 2,
43            crossAxisSpacing: 8.0,
44            mainAxisSpacing: 8.0,
45            childAspectRatio: 0.8,
46          ), // SliverGridDelegateWithFixedCrossAxisCount
47          itemCount: vm.cats.length,
48          itemBuilder: (_, index) {
49            final cat = vm.cats[index];
50            return Card(
51              elevation: 4.0,
52              child: ClipRRect(
53                borderRadius: BorderRadius.circular(8.0),
54                child: Image.network(
55                  cat.imageUrl,
56                  fit: BoxFit.cover,
57                ), // Image.network
58              ), // ClipRRect
59            ); // Card
60          },
61        ), // GridView.builder
62      ); // Scaffold
63    }
64  }
```

Captura de la App

08:01



Gatitos

